```cpp
//Task 1: Basic Input and Output .

//a
#include <iostream>
using namespace std;
int main()
{
    cout<<"Welcome to Data Structures Lab "<< endl ;
    cout<<"Course Code : BTCS209P";

    return 0;
}

//b
#include <iostream>
using namespace std;
int main()
{
    int a,b,sum,sub,multi,divd;
a = 4 ;
b= 2;
sum = a+b ;
sub = a-b ;
multi = a*b;
divd = a/b;{
cout<<"sum = "<<sum<<endl;
cout<<"subtract = "<<sub<<endl;
cout<<"multiply = "<<multi<<endl;
cout<<"division ="<<divd<<endl;
}
return 0;
}
```

```cpp
//Task 2: Decision Making Statements.

//a
#include <iostream>
using namespace std;
int main()
{
    int a ;
    cout<<"Enter a Number:";
    cin>>a;
    if ( a%2==0 )
        cout<<"Even number";

    else
        cout<<"Odd number";

    return 0;
}

//b
#include <iostream>
using namespace std;
int main()
{
    int a , b, c ;
    cout<<"Enter first no.:";
    cin>>a;
    cout<<"Enter second no.:";
    cin>>b ;
    cout<<"Enter third no.:";
    cin>>c;
    {
    if (a>b and a>c)
    cout<<"a is largest";
    else if (b>a and b>c)
```

```
    cout<<"b is largest";
    else
    cout<<"c is largest";
    }
    return 0;
}

//c
#include <iostream>
using namespace std;
int main()
{
    int y;
    cout<<"Enter year:";
    cin>>y;
    {
        if (y%4==0 && y%100!=0 || y%400==0)
        cout<<y<<" "<<"is Leap year";
        else
        cout<<y<<" "<<"is not leap year";
    }
    return 0;
}
```

```
//Task 3: Looping Statements

//a
#include <iostream>
using namespace std;
int main()
{

    for(int i=1;i<=100;i++)
    cout<<i<<" ";
}

//b
#include <iostream>
using namespace std;
int main()
{
    int n;
    cout<<"Enter the no:";
    cin>>n;
    for(int i=1;i<=10;i++)
    cout<<n<<"*"<<i<<"="<<i*n<<endl;
}
```

```
//c
#include <iostream>
using namespace std;
int main()
{
    int n , fact =1 ;
    cout<<"Enter the no:";
     cin>>n;
    for (int i=1; i<=n;i++){
    fact = fact*i;}
    cout<<"Factorial of "<<n<<":"<<endl;
    cout<<fact;
}

//d
#include <iostream>
using namespace std;
int main()
{
    int n , a = 0 , b=1 , c;
cout<<"Enter the terms:";
cin>>n;
for(int i=0;i<=n;i++){
cout<<a<<" ";
c=a+b;
a=b;
b=c;}
}
```

```cpp
//Task 4: Functions

//a
#include <iostream>
using namespace std;
int sqt(int n){
return n*n;}
int main()
{ int n;
    cout<<"Enter a No.:";
    cin>>n;
    cout<<"Square:"<<sqt(n);
}

// b
#include <iostream>
using namespace std;
int max(int a,int b){
if (a>b)
return a;
else
return b;}
int main (){
    int a,b;
    cout<<"First no.:";
    cin>>a;
    cout<<"Second no.:";
    cin>>b;
cout <<"max="<< max(a,b);}
```

```cpp
//c
#include <iostream>
using namespace std;

bool isPrime(int n) {
    if (n <= 1)
        return false;

    for (int i = 2; i < n; i++) {
        if (n % i == 0)
            return false;
    }

    return true;
}
int main() {
    int n;

    cout << "Enter a number: ";
    cin >> n;

    if (isPrime(n))
        cout << "Prime Number";
    else
        cout << "Not a Prime Number";

    return 0;
}
```

```cpp
// Task 5 = Array
#include <iostream>
using namespace std;

int main() {
    int arr[10];
    int sum = 0;
    int largest, smallest;
    int search, position = -1;
// a
    cout << "Enter 10 integers:" << endl;
    for (int i = 0; i < 10; i++) {
        cin >> arr[i];
    }

    cout << "\nArray elements are: ";
    for (int i = 0; i < 10; i++) {
        cout << arr[i] << " ";
    }

 // b
    largest = smallest = arr[0];

    for (int i = 0; i < 10; i++) {
        if (arr[i] > largest)
            largest = arr[i];

        if (arr[i] < smallest)
            smallest = arr[i];
    }

    cout << "\nLargest element = " << largest;
    cout << "\nSmallest element = " << smallest;
```

```cpp
// c
    for (int i = 0; i < 10; i++) {
        sum += arr[i];
    }

    double average = sum / 10.0;

    cout << "\nSum = " << sum;
    cout << "\nAverage = " << average;

// d
    cout << "\n\nEnter element to search: ";
    cin >> search;

    for (int i = 0; i < 10; i++) {
        if (arr[i] == search) {
            position = i;
            break;
        }
    }

    if (position != -1)
        cout << "Element found at position " << position + 1 << endl;
    else
        cout << "Element not found." << endl;

    return 0;
}
```